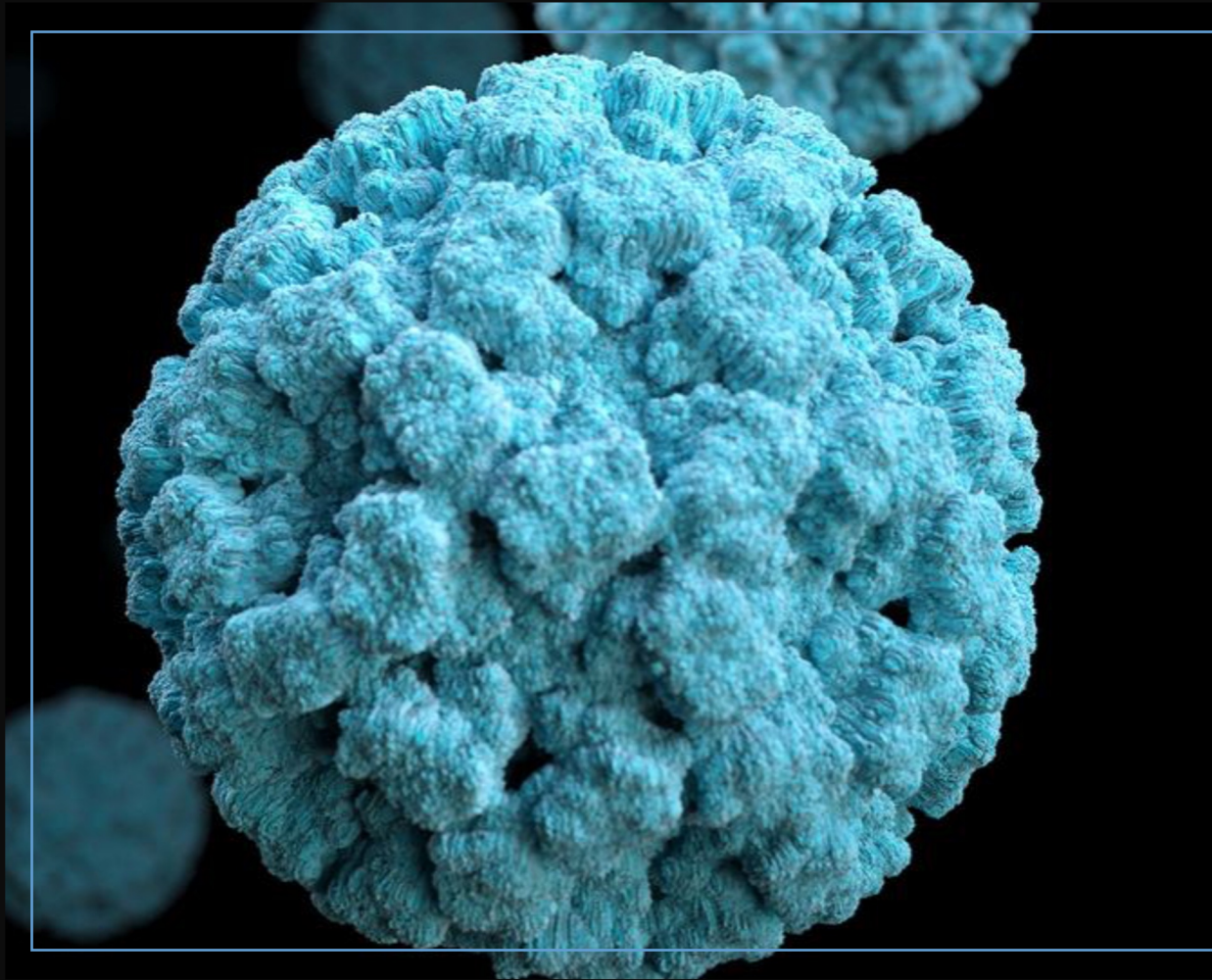




Other viruses I

Wilasinee Surasa, DVM, M.Sc.

Lecturer, Faculty of Veterinary Sciences

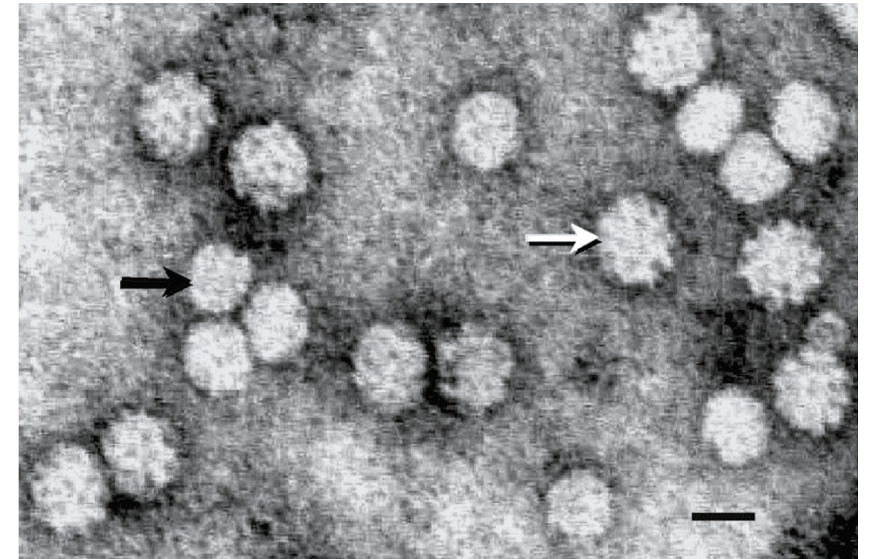
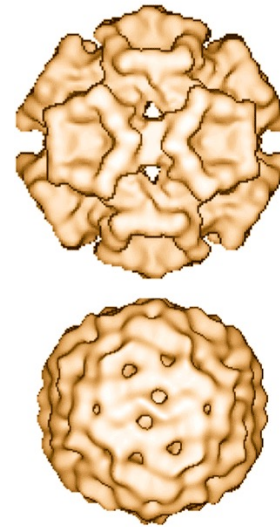
Mahasarakham University

วัตถุประสงค์

- นิสิตมีความรู้ความเข้าใจเกี่ยวกับลักษณะและชีววิทยาของไวรัส การก่อโรคในสัตว์ พยาธิกำเนิดและอาการแสดงทางคลินิก ตัวอย่างเพื่อชั้นสูตรทางห้องปฏิบัติการ วิธีการ และเทคนิคที่ใช้ชั้นสูตรทางห้องปฏิบัติการ การควบคุมและการป้องกันโรคที่เกิดจากไวรัสในตระกูลต่างๆ ดังนี้
 - Family *Circoviridae*
 - Family *Filoviridae*
 - Family *Bornaviridae*
 - Family *Arenaviridae*
 - Family *Astroviridae*
 - Family *Arteriviridae*
 - Family *Roniviridae*

Family *Circoviridae*

- Members of the family are classified in two genera –
 - Genus *Circovirus* (70 species)
 - Species include Porcine circovirus type 1 and 2 and Beak and Feather Disease virus (BFDV)
 - Some have been identified as a result of sequencing studies to characterize fecal or environmental microbiota and a few are bird pathogens
 - Genus *Cyclovirus* (90 species)
 - ICTV Taxonomy (2021) lists over 40 species associated with animals, birds, and insects
 - There is no biological data regarding the infectivity, transmission, or host range of members of the genus *Cyclovirus*



Virion properties

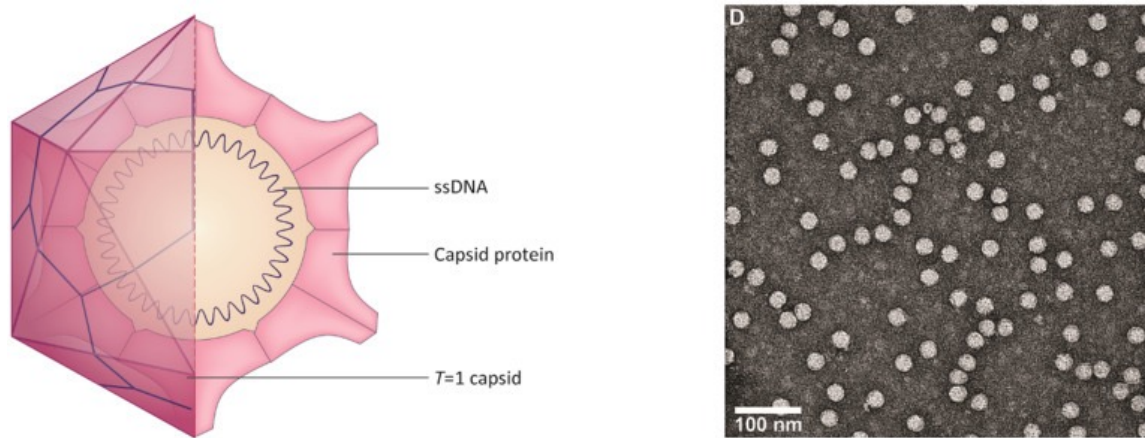
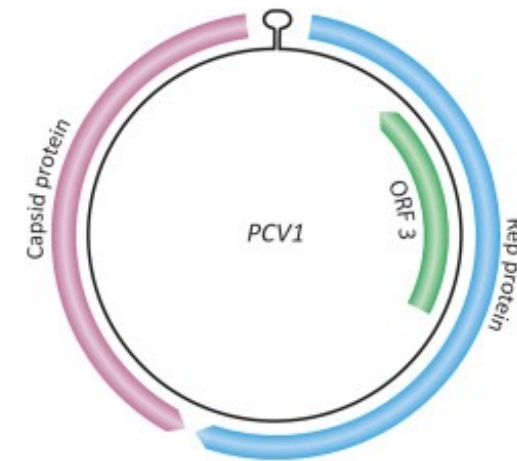


Figure: Circovirus virion structure (left) and cryo-electron microscopy of porcine circovirus 2d virus-like particles.

Source: From Khayat, R., Wen, K., Alimova, A., Gavrilov, B., Katz, A., Galarza, J.M., et al., 2019. Structural characterization of the PCV2d virus-like particle at 3.3 Å resolution reveals differences to PCV2a and PCV2b capsids, a tetranucleotide, and an N-terminus near the icosahedral 3-fold axes. *Viol.* 537, 186–197.

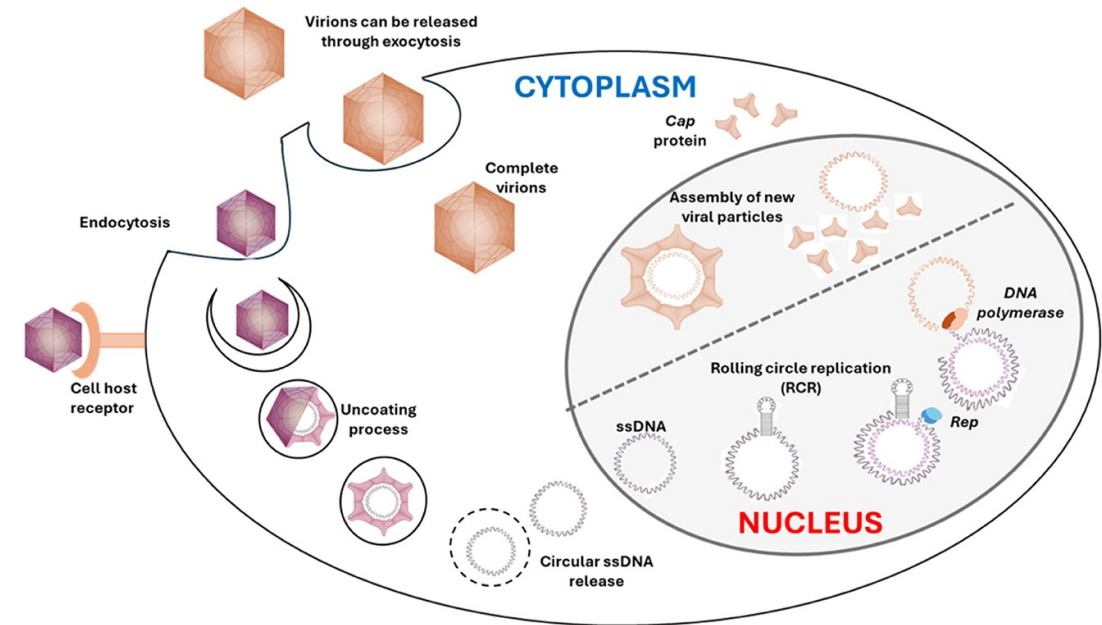
- Small (a size of ~17–25 nm diameter), non-enveloped virus
- Capsids assemble from 60 copies of the capsid (C) protein (T=1 icosahedral symmetry)
- Extremely stable in the environment
- Genome: circular, ssDNA



Genomes are covalently closed, circular, single-stranded DNA of ~1700–1800 nt with two major open-reading frames (ORFs). The genomes of members of the genus *Circovirus* are ambisense (ORFs are transcribed in opposite directions from a single promoter site). One circovirus ORF encodes the capsid (C) protein, the other encodes the nonstructural replicase-associated (Rep) protein.

Virus replication

- Replication in nucleus
 - Circoviruses replicate in mitotically active (dividing) cells as they use host cell polymerases for genome replication
 - The first step in genome replication of circoviruses is conversion of single-stranded DNA to a double-stranded molecule
 - The double-stranded genome is then transcribed by host RNA polymerase
 - Genome replication is then initiated by Rep and catalyzed by host DNA synthesis machinery
 - Detailed models for other processes in the circovirus life cycle (i.e., entry, assembly) are lacking but it is assumed that entry is mediated by endocytosis, and the virion makes its way to the nucleus

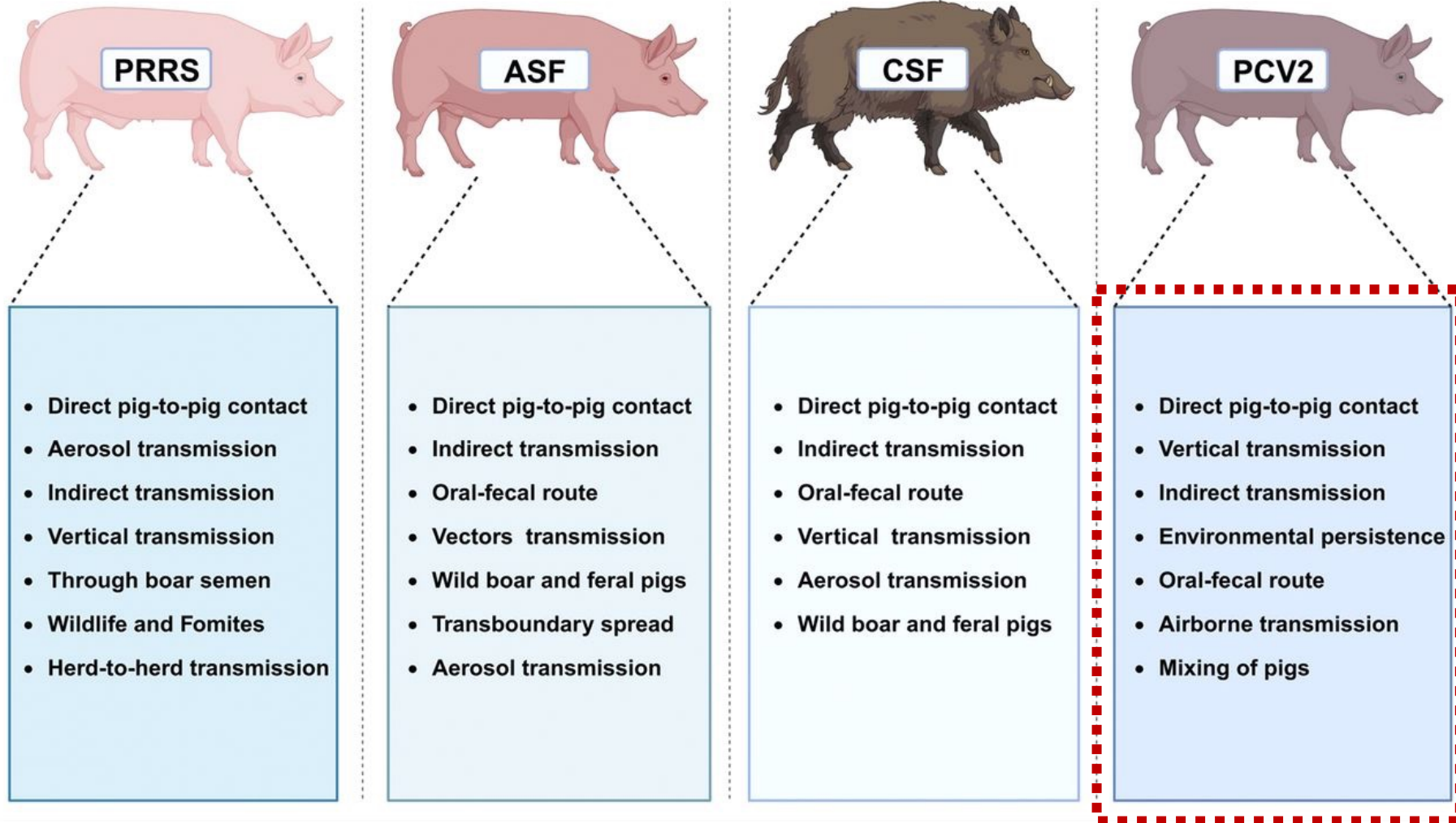


The CanineCV replication cycle

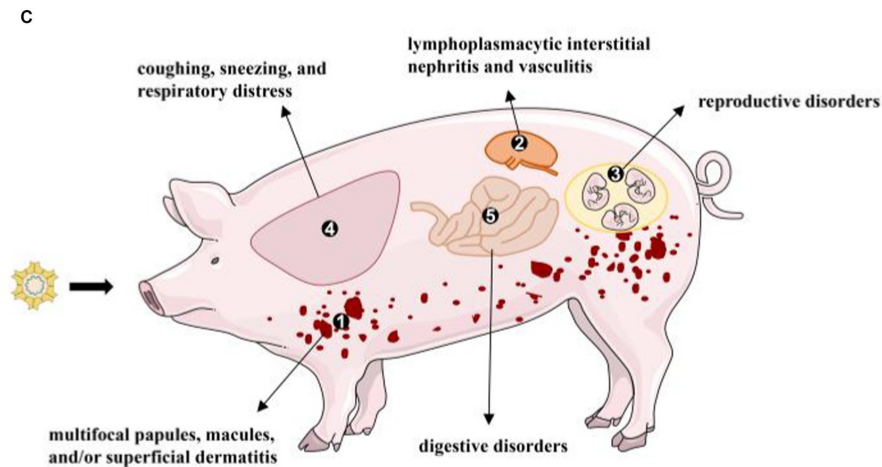
Porcine circovirus; PCV

- Include 4 types (PCV1-4):
 - PCV1: non-pathogenic strain
 - PCV2 (2a, 2b): multiple clinical presentations, such as post-weaning multisystemic wasting syndrome in swine
 - PCV3 (3a, 3b, and 3c): a new virus, causing similar symptoms to PCV2
 - PCV4: limited information about the pathogenesis and clinical implications

Transmission



Post-weaning multisystemic wasting syndrome; PMWS



- กลุ่มอาการทรุดโทรมหลังหย่านมในสุกร
- **Pathogenesis:** PCV mainly proliferates in lymph nodes and induces immune cell apoptosis to decrease the immunity of infected pigs
 - The immune response can lead to a series of serious consequences, including rapid upregulation of pro-inflammatory factors, the formation of immune complexes, decreased antigen presentation ability, and necrosis of lymphocytes and immune cells

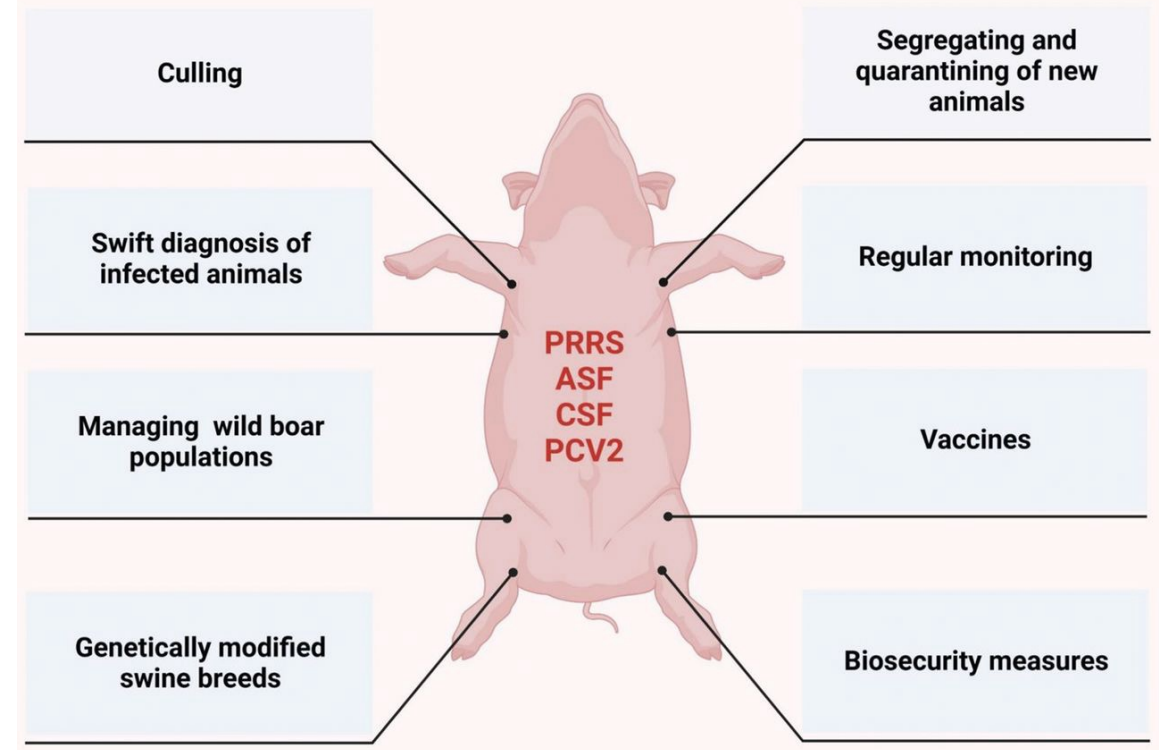
- **Clinical and pathological symptoms:** lymphadenopathy, immunosuppression
 - Skin: multifocal papules, macules, and/or superficial dermatitis
 - Kidney: lymphoplasmacytic interstitial nephritis and vasculitis
 - Reproductive disorders: abortion, stillbirth, and mummified fetuses
 - Lung: coughing, sneezing, and respiratory distress associated with lymphoplasmacytic interstitial pneumonia
 - Digestive disorders: enteritis, diarrhea, hepatitis, pancreatitis

Diagnosis and control

- Laboratory diagnosis: EM, Immunohistochemistry, PCR
- Serological test: ELISA
- Prevention and control: vaccination



Control and management



Psittacine beak and feather disease virus; PBFDV

- Primarily affects in young parrots
 - Infected birds present with abnormal feathers, beak lesions, and immunosuppression
 - The natural infection appears to occur primarily in juvenile birds, with few instances of clinical infection seen in birds >3 years old
 - This disease is still seen in African grey parrots, Eclectus parrots, lovebirds (Agapornis), lorikeets, and other species, but is rare
- Transmission: inhalation and/or ingestion, vertical transmission
 - Infected birds shed virus in their feathers, feather dander, feces, and oral secretions
 - A psittacine circovirus is very stable in the environment, so fomites are a notable source of infection

Clinical signs

- Classic PBFD infection: lack of powder down on the beak, abnormal formation of growing feathers and immunosuppression
 - Feathers are pinched or clubbed at their base and may have hemorrhage within the developing shaft
 - The feathers fall out easily and grow back slowly or not at all
 - Pigment loss may occur in colored feathers
 - The bird may live with these lesions for months to years
- Chronic disease: the immune system is affected, and most birds die of secondary infections
- A per-acute form in young birds: which develop enteritis and pneumonia, lose weight, and die



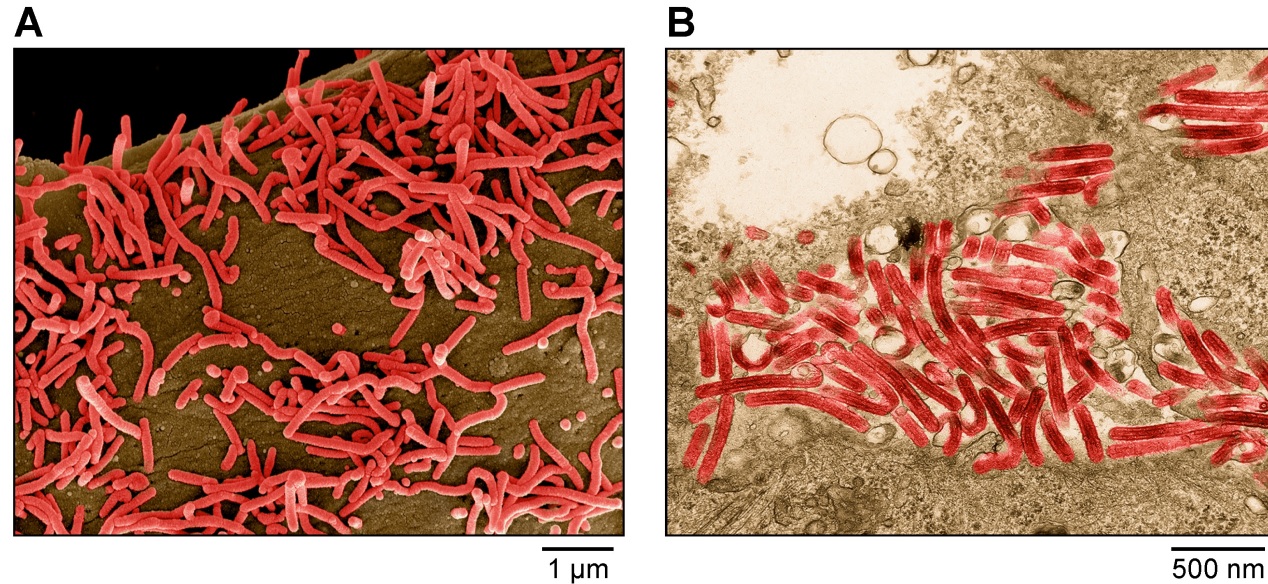
Diagnosis

- Diagnosis of PBFD is based on clinical appearance
- Laboratory diagnosis: PCR assay of feces, feather dander, or blood; and biopsy of affected feather follicles showing basophilic intracytoplasmic inclusions
 - Quarantine and retesting are recommended for PCR-positive, asymptomatic birds
- No specific gross lesions internally, but intranuclear or intracytoplasmic inclusions may be seen histologically in the feathers, bursa, thymus, liver, or other organs

Treatment and control

- No specific treatment for PBFD, and treatment of infected birds is supportive
 - The clinically affected birds warrant isolation and eventual euthanasia in most clinical cases
- Strict hygiene with attention to dust control, screening protocols including PCR testing of both birds and the environment, and lengthy quarantines are highly recommended in breeding facilities with susceptible species

Family *Filoviridae*

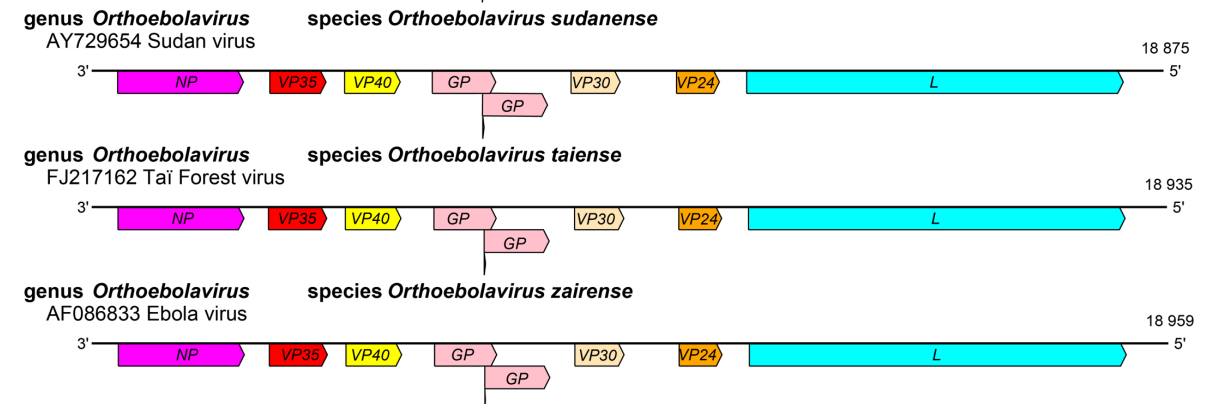
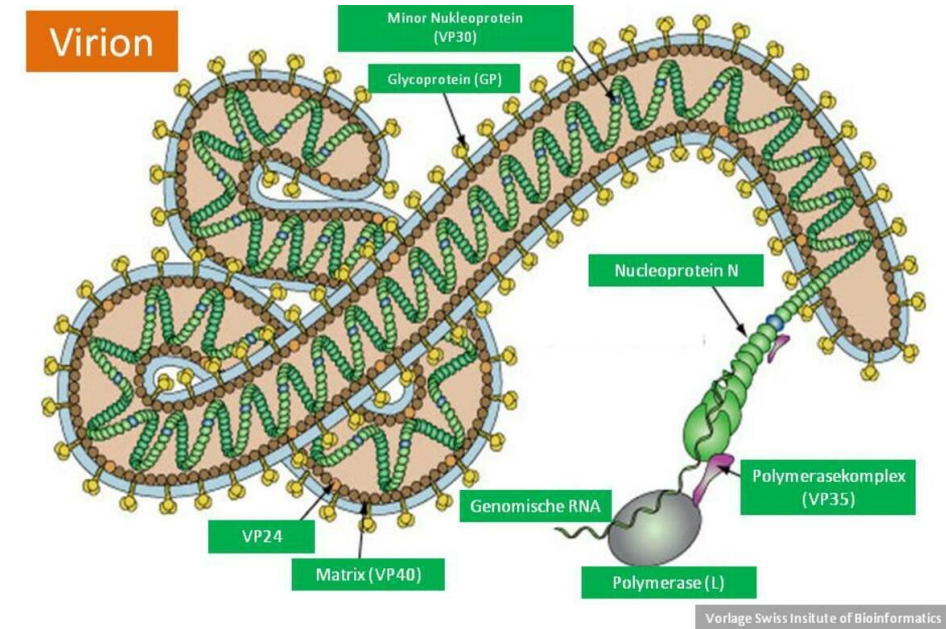


***Filoviridae*.** A) Scanning electron micrograph of Marburg virus particles (red) budding from an infected grivet (*Chlorocebus aethiops* (Linnaeus, 1758)) Vero E6 cell. B) Transmission electron micrograph of Marburg virus particles (red) found both as extracellular particles and budding particles from Vero E6 cells. Images are colorized for clarity. Courtesy of John G. Bernbaum and Jiro Wada, NIH/NIAID/DCR/IRF Frederick, Fort Detrick, MD, USA.

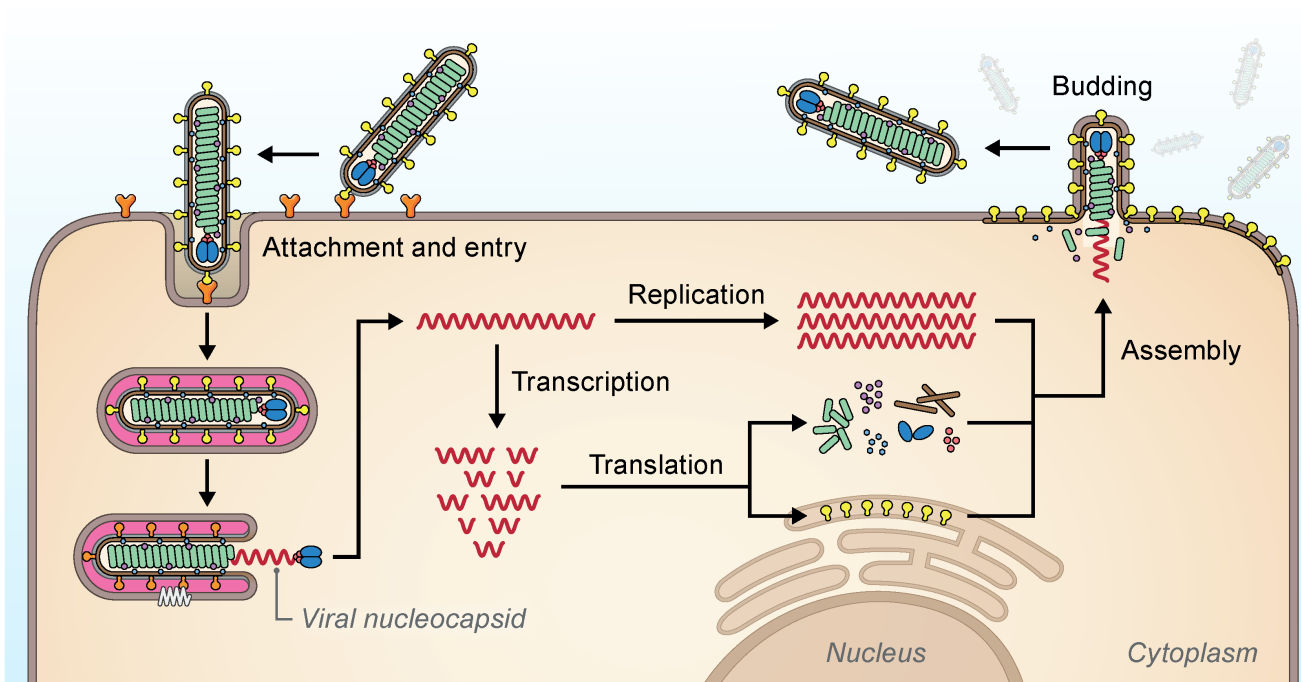
- The family includes 8 genera –
 - Piscine hosts
 - Genus *Oblavirus*
 - Genus *Striavirus*
 - Genus *Thamnovirus*
 - Mammalian hosts
 - Genus *Cuevavirus*
 - Genus *Dianlovirus*
 - Genus *Orthoebolavirus*
 - Genus *Orthomarburgvirus*
 - Reptilian hosts
 - Genus *Tapjovirus*

Virion properties

- Enveloped, variously shaped but predominantly filamentous (diameter is about 80 nm, 790 nm long for Marburg virus and 970 nm long for Ebolavirus), typically, with a single nucleocapsid
- Genome: 13.1–20.9 kb of linear, negative-sense, non-segmented RNA
 - Encodes for seven proteins: nucleoprotein (NP), polymerase complex (VP35), matrix protein (VP40), glycoprotein (GP), minor nucleoprotein (VP30), VP24, and polymerase (L)
- Host range: fish, mammals, and reptiles



Virus replication



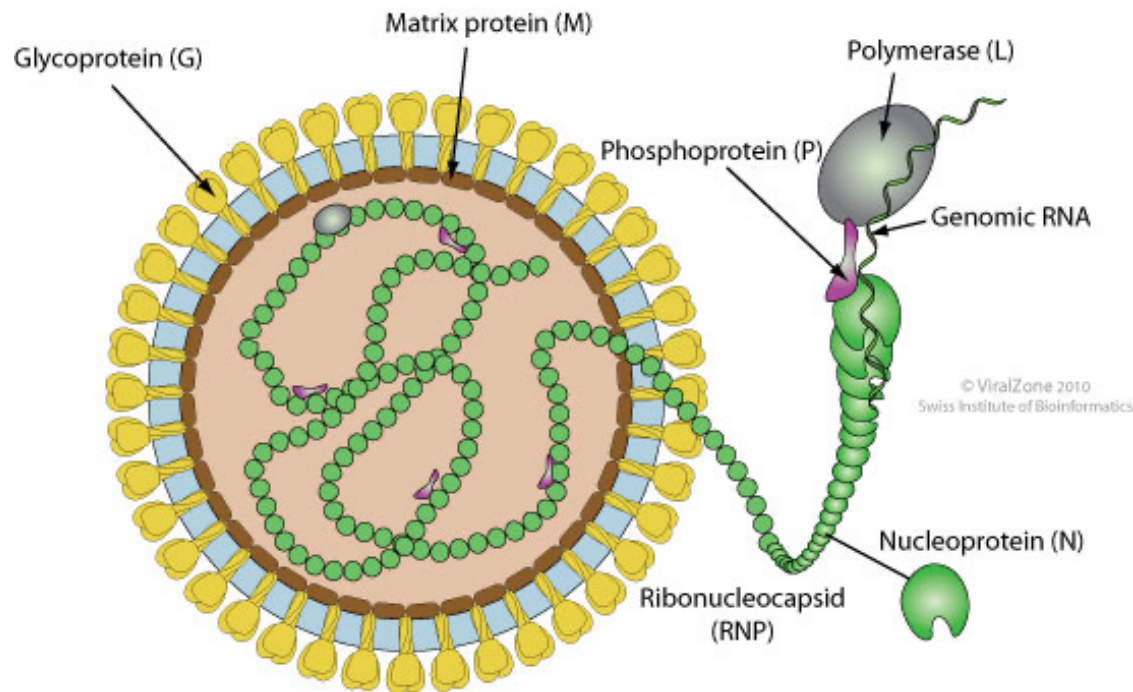
- Cytoplasmic replication
 - Virions attach to cell-surface attachment factors and are taken into the cell via endocytosis
 - The filovirion glycoproteins bind to endosomal NPC intracellular cholesterol transporter 1 (NPC1) and catalyze the fusion of viral and cellular membranes to release the filovirid RNP complex
 - The polymerase complex (consisting of VP35 and L) transcribes filovirid mRNAs, which are translated into filovirid proteins, and replicates filovirid genomic RNA via antigenomic intermediates
 - Assembly of filovirid proteins and progeny genomes occurs in the cytoplasm and results in budding and release of virions at the plasma membrane

Filoviruses infection

- **Viral hemorrhagic fever – human and non-human primate**
 - Genus *Cuevavirus*: Lloviu virus (LLOV), discovered in dead miniopterid bats (likely incidental hosts) - reported from Europe
 - Genus *Dianlovirus*: Měnglà virus (MLAV), discovered in pteropodid bats - reported from China
 - Genus *Ebolavirus*: includes six species for six viruses
 - **Bombali virus (BOMV)** - detected in molossid bats
 - **Ebola virus (EBOV)** and **Reston virus (RESTV)**, are suspected to be harbored by bats as natural hosts
 - RESTV has also been found in domestic pigs
 - 5 ebolaviruses includes **Bundibugyo virus (BDBV)**, EBOV, RESTV, **Sudan virus (SUDV)**, and **Tai Forest virus (TAFV)** are pathogenic for non-human primates
 - BDBV, EBOV, and SUDV are highly lethal human pathogens
 - Genus *Marburgvirus*: Marburg virus (MARV) and Ravn virus (RAVV)
 - Highly lethal human pathogens that are endemic in Africa



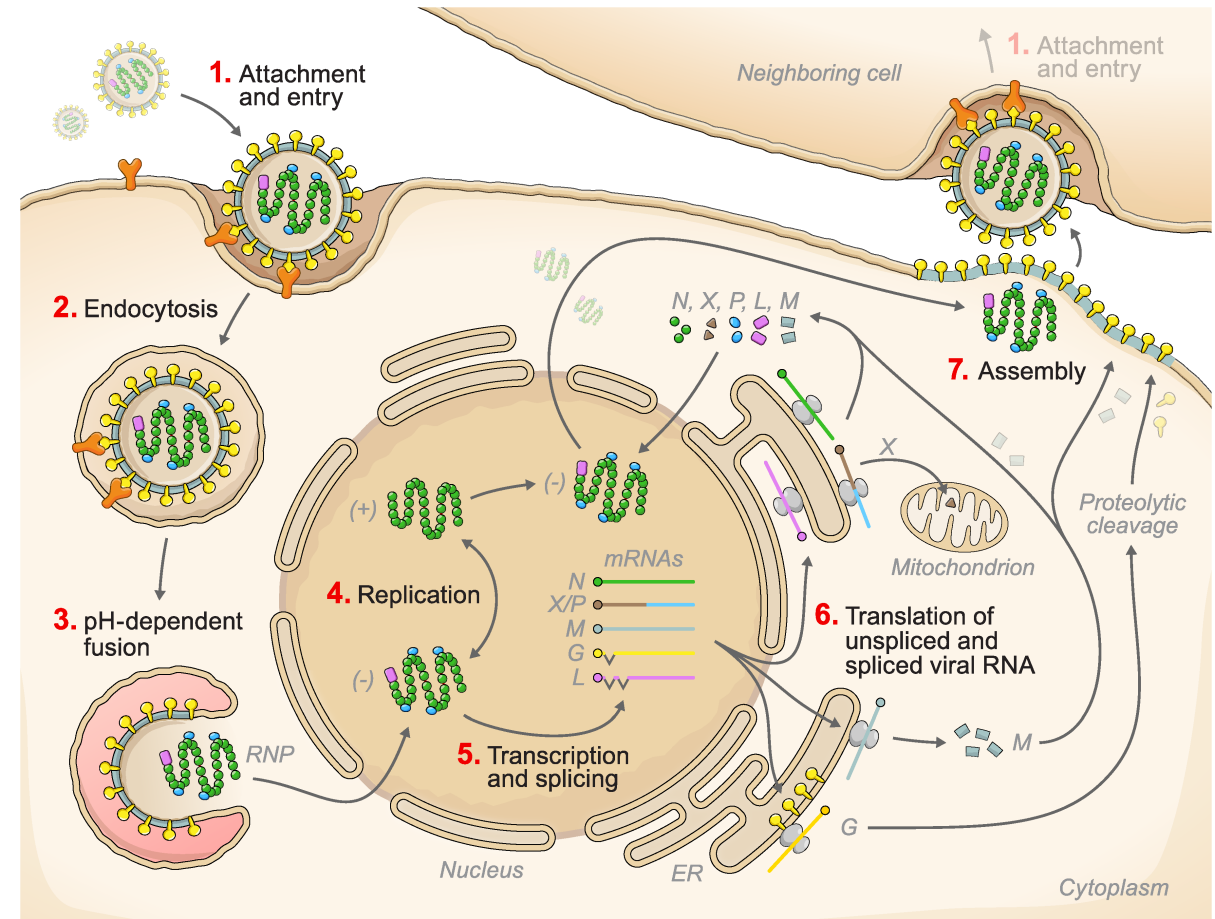
Family *Bornaviridae*



- Genus *Bornavirus*; Borna disease virus (BDV)
- Enveloped, spherical, diameter from 70-130 nm
- Genome: linear, negative-sense ssRNA
 - Encode 6 proteins (N, P, X, M, G, and L)
- การทำลายไวรัส:
 - Heat treatment at 56°C
 - Inactivated below pH 5.0
 - Organic solvents, detergents, and exposure to UV radiation
 - Chlorine-containing disinfectants or formaldehyde

Virus replication

- ไวรัสเพิ่มจำนวนใน nucleus
- เข้าสู่ host cell โดยกระบวนการ endocytosis
- ออกจาก host cell โดยการ budding

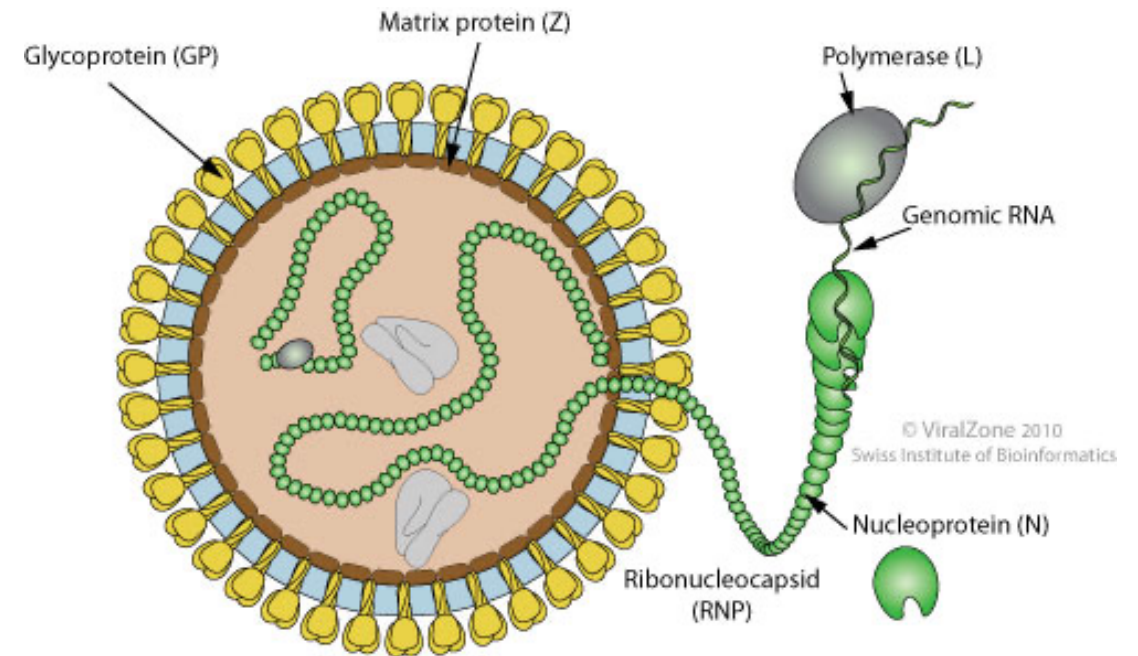


Borna disease virus (BDV)

- Transmission: fomites, direct contact with salivary, urine and feces
- BDV; cause a fatal neurologic disease called Borna disease (BD) – **Zoonosis!**
 - Associated fatal human encephalitis
 - Horses and sheep - the main natural hosts of BDV
 - Neonatal rats - persistently infected and immunotolerant after experimental infection
 - They display developmental, emotional, and cognitive impairments.
 - Parrot - severe neurological problems including incoordination, seizures, and blindness

Family *Arenaviridae*

- Enveloped, pleomorphic virion (arenosus=sandy), 110-130 nm
- Genome: two or three linear ambi-sense or negative-sense RNA segments
- Replication in cytoplasm
- ทำให้เกิด lymphocytic choriomeningitis in mouse, rat, hamster, monkey, human





Arenaviruses infection

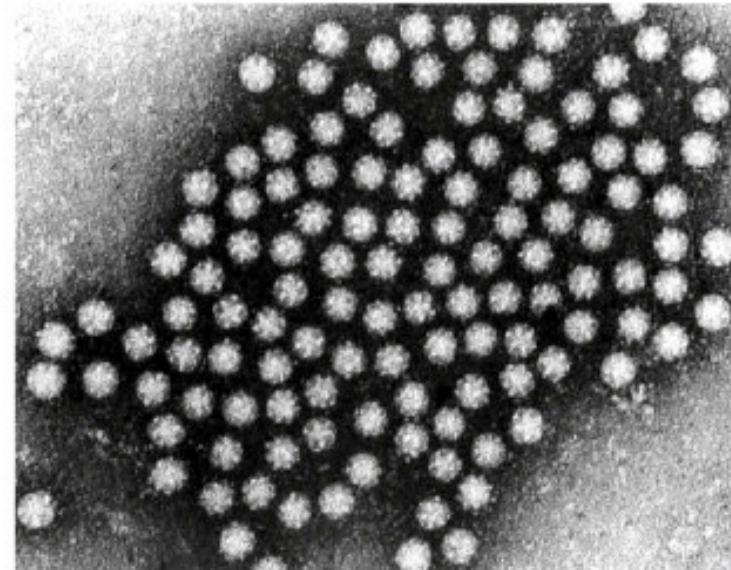
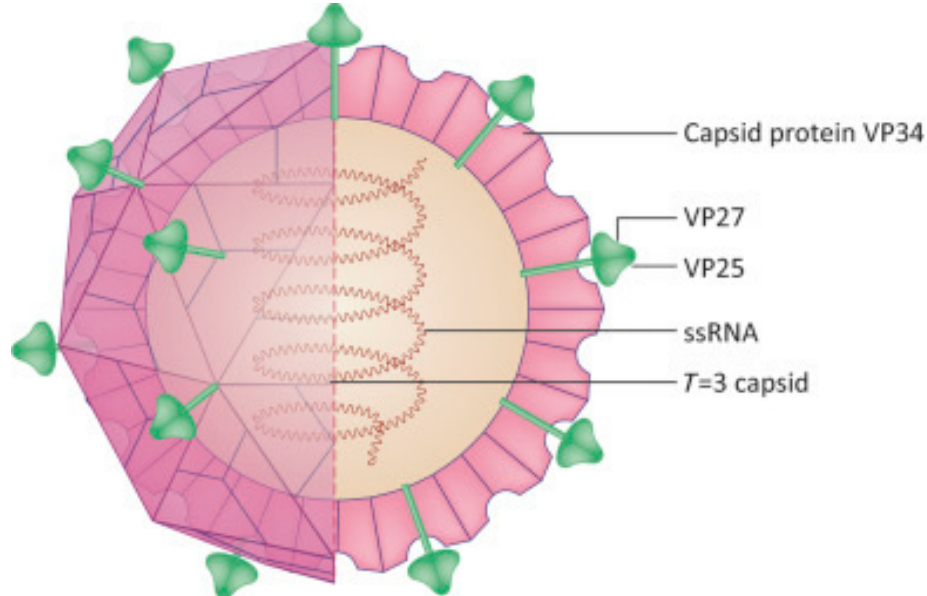
- Junin virus (Argentina)
- Machupo virus (Bolivia)
- Guanarito virus (Venezuela)
- Lassa virus (Nigeria)

**Hemorrhagic fever
(Leakage syndrome)**

- **Laboratory diagnosis:** virus isolation in cell culture, FA
- **Serological test:** ELISA

Family *Astroviridae*

- Astron = star
- Icosahedral symmetry, 40–45 nm in diameter
- Genome: unsegmented, positive-sense RNA with ~7–9 kb
- Non-enveloped; spikes that project about 41 nm from the surface of the capsid, giving them a star-like appearance
- Replication is cytoplasm

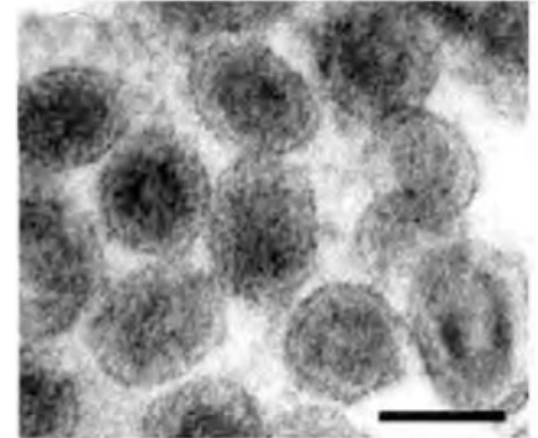
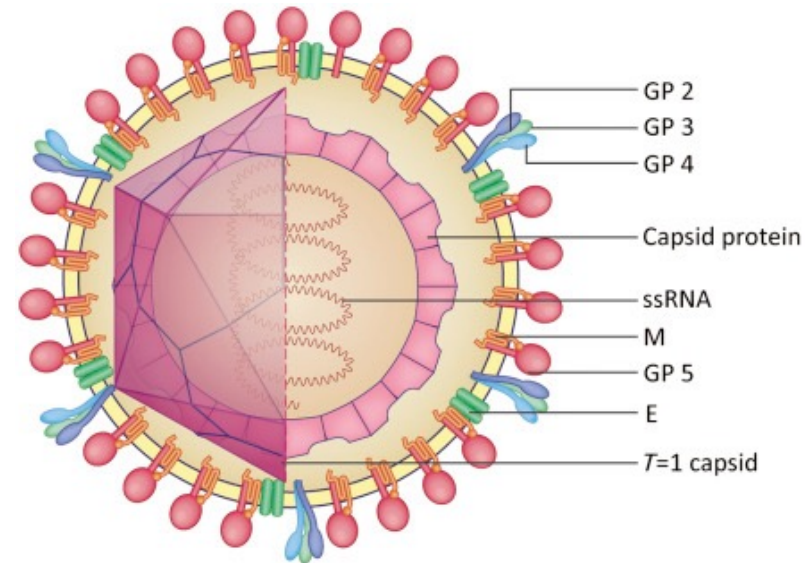


Astroviruses infection

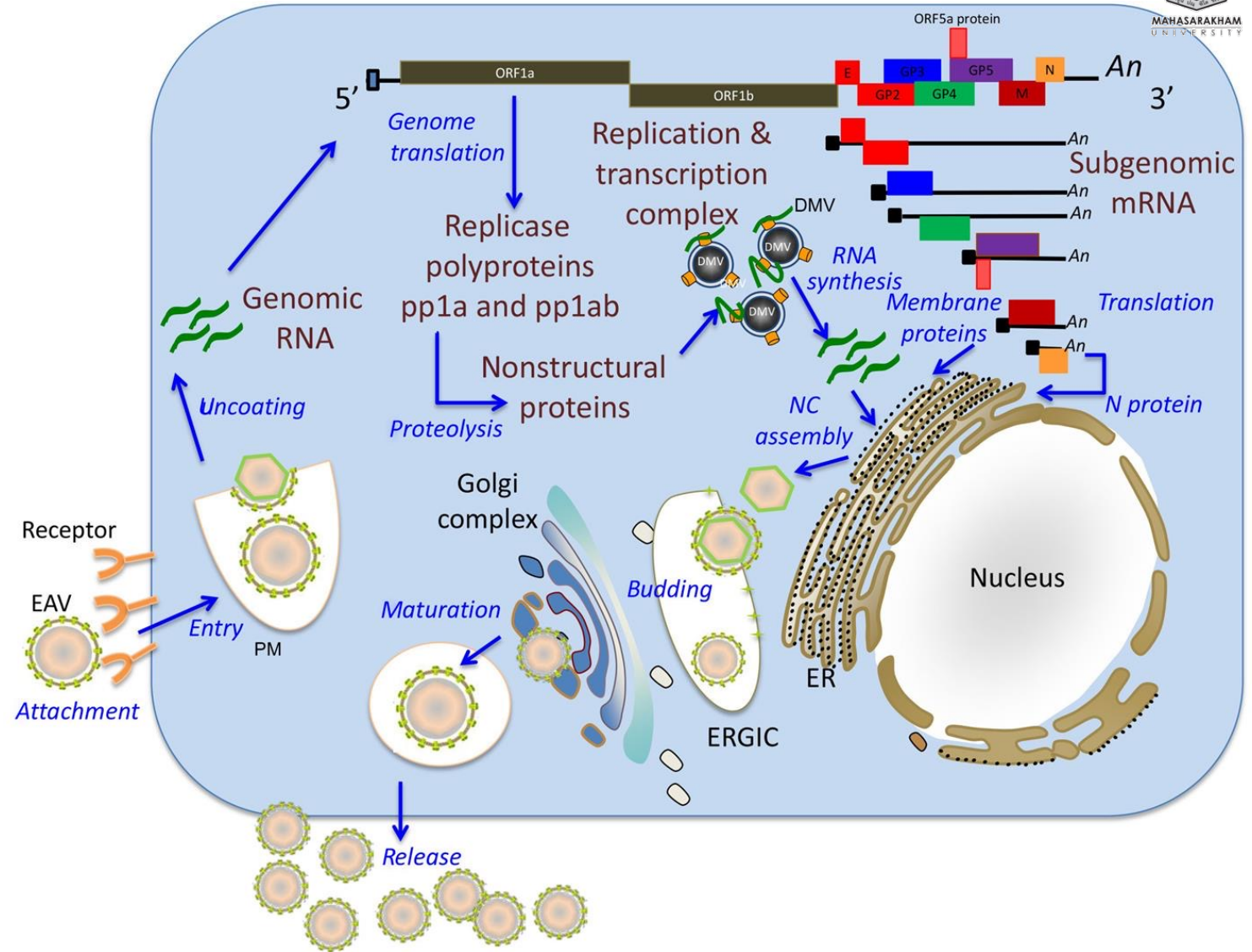
- Human astroviruses cause **gastroenteritis in children and adults**
 - Most common cause of viral diarrhea in young children
 - Sporadic outbreaks of acute gastroenteritis in adults
 - Transmission by contaminated food and water
 - Symptoms: diarrhea, nausea, vomiting, fever, malaise, and abdominal pain; for the most part disease is self-limiting
- The astroviruses have been isolated from many other mammals including
 - Pigs, cats, minks, dogs, rats, bats, calves, sheep, and deer
 - Marine mammals such as sea lions and dolphins
 - Birds; they can cause significant disease in turkeys, ducks, and chickens

Family *Arteriviridae*

- **Virion properties**
 - Spherical, enveloped with icosahedral capsid, 60-70 nm
 - Positive linear-ssRNA
 - Replicate in cytoplasm



Virus replication



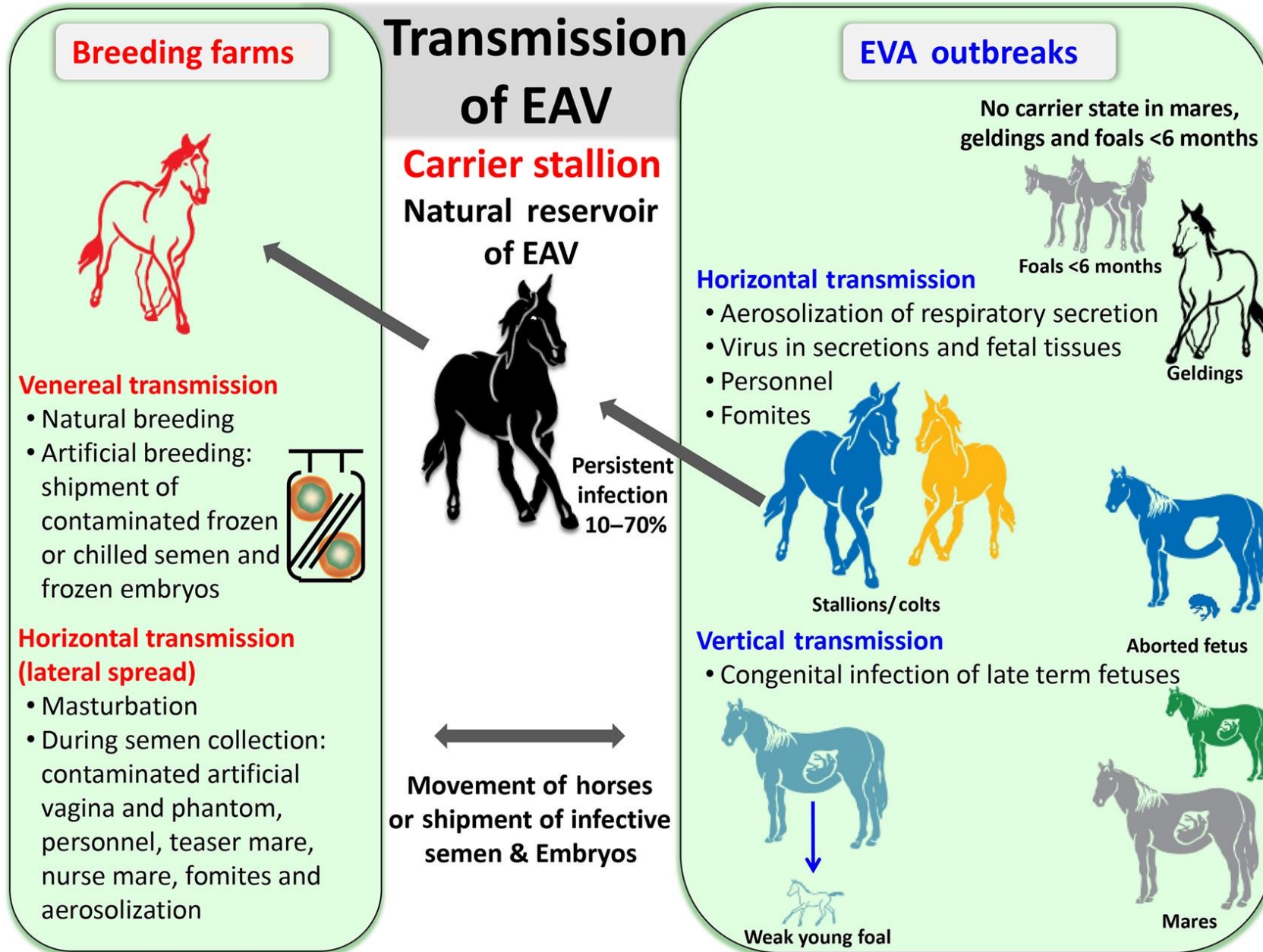
Arterivirus infection in animals

- Equine arteritis virus
 - Respiratory disease, congenital disease, generalized infection, abortion
- Simian hemorrhagic fever virus
 - Hemorrhagic fever
- Lelystad virus
 - Respiratory disease, infertility, congenital disease



Equine arteritis virus

- ก่อโรค *Equine arteritis* ในม้า ลา ล่อ
- ลักษณะเด่น: Pink eye, epidermic cellulitis
- **Transmission:** contact, aerosol, aborted fetus and placenta
- **พยาธิสภาพ:** ทำให้เกิด segmental necrosis of small artery ทั่วตัว



Clinical signs, diagnosis and control

- **Clinical signs:**
 - Fever, anorexia, depressed, leukopenia, conjunctivitis with edema of the conjunctiva, edema of the legs and ventral abdominal wall, abortion
- การวินิจฉัย: ELISA, RT-PCR
- การป้องกัน: มีการใช้วัคซีนที่เตรียมในเซลล์เพาะเลี้ยงในม้าแม่พันธุ์ราคาแพง



Simian hemorrhagic fever virus



- ก่อโรค Simian hemorrhagic fever พบครั้งแรกในลิงที่นำเข้าจากอินเดีย (เป็นไขเลือดออก)
- ก่อโรคใน Macaques (*Macaca fascicularis*) old world monkey อัตราการตายมากถึง 100%
- **Transmission:** contact, aerosol, fomite
- **Clinical signs:** fever, anorexia, facial edema diarrhea dehydration, hemorrhage
- **Lesions:** พบจุดเลือดออกใน dermis, เยื่อบุเมือกที่ช่องจมูก ปอด ลำไส้และอวัยวะในช่องท้อง

Lelystad virus



- Porcine reproductive and respiratory syndrome (PRRS)
- Mystery Swine Disease (MSD), Blue ear disease, Swine Infertility and Respiratory Syndrome (SIRS), Porcine Epidemic Abortion and Respiratory Syndrome (PEARS)
- **Clinical signs:** ปั่นสีม่วงน้ำเงินที่บริเวณหู ท้อง, abortion, ตายแรกคลอด, mummification



Porcine reproductive and respiratory syndrome (PRRS)

- การแพร่กระจายของเชื้อก่อโรค
 - การนำสุกรป่วยหรือสุกรที่เป็นตัวกักโรคเข้าสู่ฝูง
 - การแพร่กระจายของเชื้อทางอากาศภายในรัศมี 3 กิโลเมตร (โดยทั่วไปประมาณ 2 กิโลเมตร)
 - การสัมผัสโดยตรงกับสัตว์ป่วย เชื้อไวรัสจำนวนมากจะถูกขับออกมาทางการหายใจและอุจจาระ เชื้อจึงแพร่จากสุกรป่วยไปยังสุกรอื่นได้ง่าย โดยเฉพาะจากการสูดดมและเลียกัน
 - การใช้น้ำเชื้อที่ผสมเทียมจากพ่อพันธุ์ที่มีการติดเชื้อมีการติดต่อผ่านน้ำเชื้อสุกร

Clinical signs, diagnosis and control

- Clinical signs

- สุกรแม่พันธุ์ พบว่ามีการคลอดก่อนกำหนด แท้งในระยะท้ายของการตั้งท้อง (มากกว่า 100 วัน) ลูกที่คลอดอ่อนแอ อัตราการเกิดมัมมี่และลูกตายแรกคลอดสูง
- สุกรดุนม สุกรอนุบาล สุกรขุน มักมีปัญหาเกี่ยวกับระบบทางเดินหายใจ แคระแกร็น โตช้า และมักพบโรคแทรกซ้อนอื่นร่วมด้วย ความรุนแรงของโรคจะลดลงเมื่อสุกรอายุมากขึ้น สุกรที่อายุมากกว่า 1 เดือน จะแสดงอาการไม่เด่นชัดถ้าไม่มีโรคแทรกซ้อน

- Diagnosis

- การเพาะแยกเชื้อ: แยกเชื้อจากปอด ซีรัม ทอลซิล เจริญได้ในเซลล์เพาะเลี้ยง swine alveolar macrophage
- Serology tests: FA, ELISA

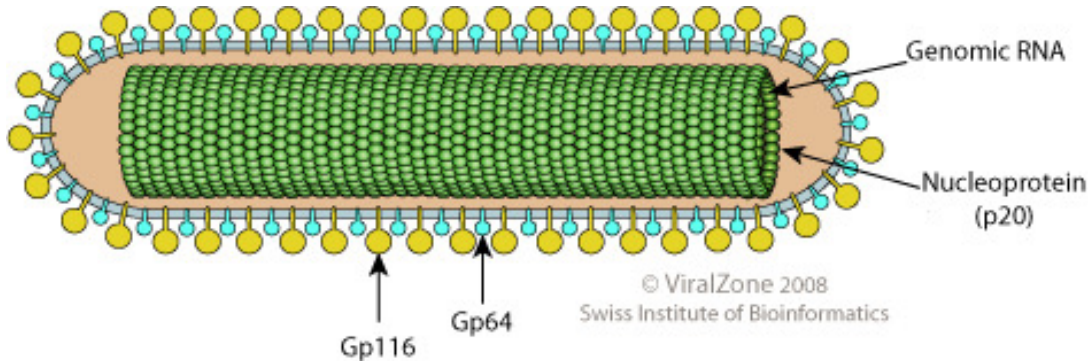
- Control

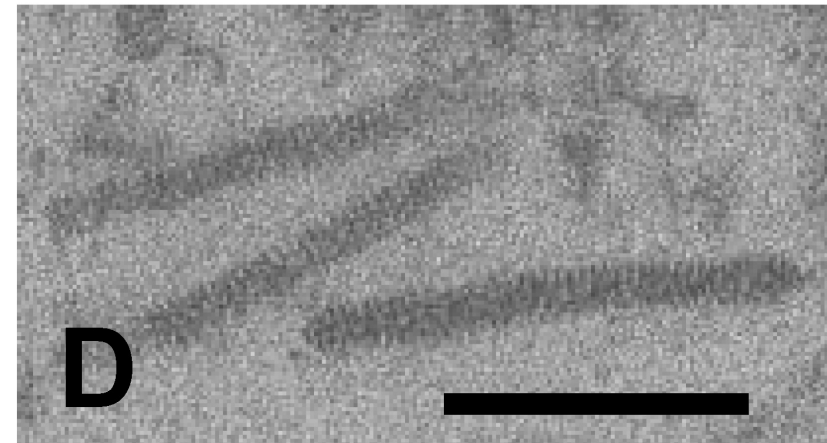
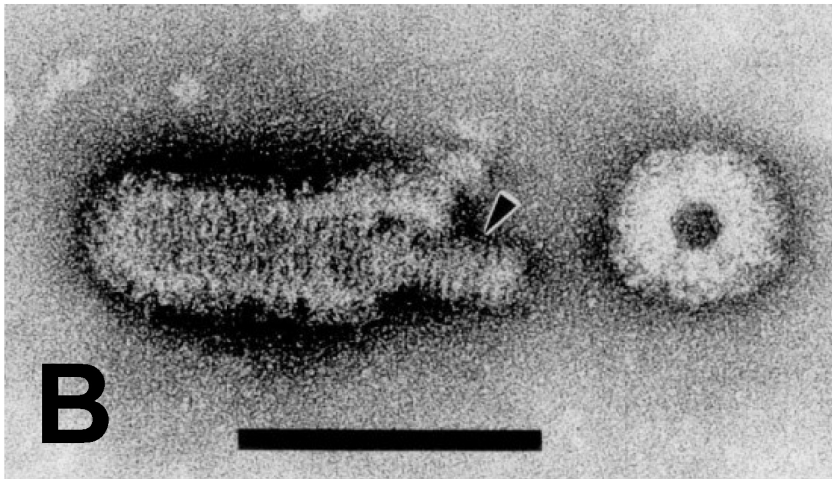
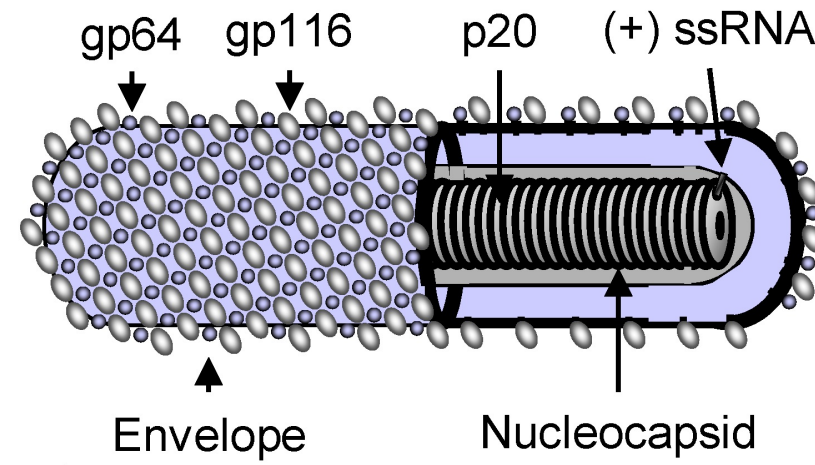
- วัคซีนเชื้อเป็นให้ผลลดระดับไวรัสและระยะของภาวะ viremia ดีกว่าวัคซีนเชื้อตาย



Family *Roniviridae*

- The family *Roniviridae* includes a single genus (*Okavirus*) for three species of viruses
- Enveloped, bacilliform-shaped
- Size: about 150-200 nm x 40-60 nm
- Nucleocapsid have helical symmetry with diameter of 20-30nm
- Genome: linear ssRNA(+) genome of 26 kb in size
- Replication: cytoplasmic; nucleocapsids bud at membranes of the endoplasmic reticulum/Golgi complex to form mature virions





Genus *Okavirus*: Yellow head virus (YHV)

- พบในกุ้งสกุล *Penaeid* หลายชนิด เช่น
 - กุ้งกุลาดำ (*P. monodon*)
 - กุ้งขาว (*P. vannamei*, *P. japonicus*, *P. setiferus*, *P. aztecus*, *P. duorarum*, *P. stylirostris*)
- **ลักษณะอาการ:** กุ้งลำตัวซีด เหงือกและบริเวณตับและตับอ่อนมีสีเหลืองเห็นชัดเจน กุ้งกินอาหารเพิ่มมาก ผิดปกติจากนั้นจะเริ่มกินลดลง เริ่มแสดงอาการหัวเหลือง ตายเร็วมาก ภายใน 3-5 วัน
- **การติดต่อ:** ติดต่อผ่านทางน้ำ อาหาร การสัมผัสโดยตรงกับไวรัส และพาหะนำเชื้อ เช่น กุ้ง ปู นก
- **การป้องกัน:** มีการจัดการภายในฟาร์มที่ดี กำจัดพาหะต่าง ๆ มีการพักน้ำเป็นเวลานานในบ่อพัก และใช้ระบบกรองอย่างดีเมื่อมีการสูบน้ำเข้าสู่บ่อเลี้ยง



References

- Carter, J.B. and Saunders, V.A. 2007. Virology: Principles and Applications. John Wiley & Sons Ltd., England. 358p.
- Hirsh, D.C., MacLachlan, N.J. and Walker, R.L. 2004. Veterinary Microbiology. 2nd edition. Blackwell Publishing. Ames, Iowa. 536p.
- MacLachlan, N.J. and Dubovi, E.J. 2011. Fenner's Veterinary Virology. 4th edition. Elsevier Inc., UK. 507p.
- Mahy, B.W.J. and Van Regenmortel, M.H.V. 2010. Desk Encyclopedia of Animal and Bacterial Virology. Elsevier Inc., UK. 617p.
- Quinn, P.J., Markey, B.K., Leonard, F.C., FitzPatrick, E.S., Fanning, S. and Hartigan, P.J. 2011. Veterinary Microbiology and Microbial Disease. 2nd edition. Wiley-Blackwell, West Sussex, UK. 912p.